Lemma. Let F be a field, $f(x) \in F[x]$ be a polynomial and $D_x f(x)$ be the derivative.
α is a multiple root of f if and only if α is a root of both f and $D_x f$.

In particular,

$f(x)$ is separable if and only if $f(x)$ and $D_x f(x)$ are relatively prime.

Proof. Let K be the splitting of $f(x)$ and $D_x f(x)$.
Suppose that f has α as a multiple root. That is, $f(x)$ is divided by $(x-\alpha)^2$, so

$$f(x) = (x-\alpha)^2 \cdot g(x) \text{ for some } g \in K[x].$$

Taking the formal derivative to f, then

$$D_x f(x) = 2 \cdot (x-\alpha) \cdot g(x) + (x-\alpha) \cdot D_x g(x)$$

Clearly it has α as a root. Conversely, $\alpha \in K$ is root of f and $D_x f$.
Then, we can represent $f(x) = (x-\alpha) \cdot h(x)$, and taking derivative then

$$D_x f(x) = h(x) + (x-\alpha) \cdot D_x h(x).$$

Since $D_x f(\alpha) = 0$ by the assumption, so $h(\alpha) = 0$. It is proved the first assertion.
Now, the second assertion is shown directly;

$f(x)$ is separable iff $f(x)$ has no multiple root iff $f(x)$ and $D_x f(x)$ are relatively prime.

Note; To clarify understand, consider the negation. ∎

Corollary. Every irreducible polynomial over a field of characteristic 0 is separable.
Proof. Let F be a field of characteristic 0, and $f(x) \in F[x]$ be an irreducible.
Then, $D_x f(x)$ is polynomial of degree $\deg f - 1$, so therefore it must be relatively prime. ∎

Note; But, in the field of characteristic $p \neq 0$, $D_x f(x)$ can be zero, Thm. A field of characteristic 0 is perfect.

Proof. Let $f(x) \in F[x]$ be an irreducible polynomial. If $D_x f(x)$ is not zero, then

$$\deg D_x f < \deg f.$$

Therefore, $(f, D_x f) = 1$, because if $(f, D_x f) \neq 1$, then it is contradiction to f is irreducible..
So it is separable.

Lemma. A field of characteristic p is perfect iff every element of F has a pth root in F.

Proof. Let F be a field of characteristic p.
Suppose that F is perfect. Let $\alpha \in F$ be a non-zero, non-1. Put $f(x) = x^p - \alpha$.
If $f(x)$ has a root in F, then there is nothing to prove.
Assume f has no root in F. Observe that in splitting field K of $f(x)$, put $\beta \in K$ s.t.

$$f(\beta) = \beta^p - \alpha = 0.$$

And, $f(x) = x^p - \alpha = x^p - \beta^p = (x-\beta)^p$ in K. But, if f is reducible in F, then for some $1 \le k < p$, $f(x) = (x-\beta)^{p-k} \cdot (x-\beta)^k$ and $(x-\beta)^{p-k}, (x-\beta)^k \in F[x]$.

That is, $(-\beta)^k \in F$. Clearly, $(k, p) = 1$, thus $ku + pv = 1$ for some $u, v \in \mathbb{Z}$.
Now, $\beta = \beta^{ku+pv} = (\beta^k)^u \cdot (\beta^p)^v = (\beta^k)^u \cdot \alpha^v \in F$. It is contradiction. But, if f is irreducible then $D_x f = p \cdot x^{p-1} = 0$ implies contradiction with F is perfect. Now, α has a p'th root in F.
Conversely, let $f(x) \in F[x]$ be an irreducible. Assume it is inseparable.
Denote $f(x) = a_n x^n + \cdots + a_1 x + a_0$. Since $f(x)$ is irreducible and inseparable,
$D_x f(x) = n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \cdots + a_1$ must be zero, that is,

$$a_k \neq 0 \iff k \in (p)$$

Therefore, we can write

$$f(x) = b_m \cdot x^{mp} + b_{m-1} x^{(m-1)p} + \cdots + b_1 x^p + b_0$$

Now by the assumption, each b_i has p'th root c_i, thus

$$f(x) = c_m^p \cdot (x^m)^p + c_{m-1}^p \cdot (x^{m-1})^p + \cdots + c_0^p$$

$$= (c_m \cdot x^m + c_{m-1} \cdot x^{m-1} + \cdots + c_0)^p.$$ But it is reducible. Contradiction.

Corollary. Every finite field is perfect. ($a \mapsto a^p$ is 1-1 on finite, thus Auto.)

「Thm. Every finite separable extension is simple extension.

Proof. If F is finite, then the finite extension E/F is also finite,

Then, $E^{\times}$ is cyclic, therefore $E = F(\alpha)$ where α is generator of $E^{\times}$.

Now, assume that F is infinite. Since E/F is finite,

$$E = F(\alpha_1, \cdots, \alpha_r) \quad \text{where } \alpha_1, \cdots, \alpha_r \in E.$$

Now, enough to show $F(\beta, \gamma) = F(\alpha)$ for some $\alpha \in E$, and after is solved by the induction.

Let $\beta_1, \cdots, \beta_n$ be distinct zeros of $\mathrm{irr}(\beta, F)$, and

$\gamma_1, \cdots, \gamma_m$ be distinct zeros of $\mathrm{irr}(\gamma, F)$, in $\overline{F}$. $(\beta_1 = \beta, \ \gamma_1 = \gamma)$.

Since E/F is separable, each zeros are multiplicity of 1.

Take $a \in F$ s.t. $a \neq \dfrac{\beta_i - \beta}{\gamma - \gamma_j}$ for all $1 \le i \le n$, $2 \le j \le m$, since F is infinite.

Put $\alpha = \beta + a \cdot \gamma$. Then, for all $(i, j) \in \{1, \cdots, n\} \times \{2, \cdots, m\}$,

$\alpha = \beta + a\gamma \neq \beta_i + a \cdot \gamma_j$, or $\alpha - a \cdot \gamma_j \neq \beta_i$.

observe that: $\mathrm{irr}(\beta, F)(\alpha - ax) \in F(\alpha)[x]$, and

$\mathrm{irr}(\beta, F)(\alpha - a\gamma) = \mathrm{irr}(\beta, F)(\beta) = 0$.

But, $\mathrm{irr}(\beta, F)(\alpha - a \cdot \gamma_i) \neq 0$ for all $2 \le i \le m$,

That is, $\mathrm{irr}(\beta, F)(\alpha - ax)$ and $\mathrm{irr}(\beta, F)(x)$ have a common factor $(x - \gamma)$ in $F(\alpha)$,

thus $\gamma \in F(\alpha)$, so $\beta = \alpha - a\gamma \in F(\alpha)$. ∎